MILITARY SPECIFICATION

BROILER-TOASTER-GRIDDLES, GAS FIRED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a gas-fired broiler-toaster-griddle.

1.2 Classification. The broiler-toaster-griddle covered by this specification shall be of the following grades as specified (see 6.2).

Grade A - Stainless steel
Grade B - Carbon steel, with manufacturers' commercial finish

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-R-11337 - Ranges, Packaging and Packing Of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 7310

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 4 - Commercial Cooking and Hot Food Storage Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip

A 176 - Stainless and Heat-Resisting Chromium-Steel Plate, Sheet and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

Z21.3 - Standard for Hotel and Restaurant Gas Ranges and Unit Broilers

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018; or the American Gas Association, 8501 East Pleasant Valley Road, Cleveland, OH 44131.)

AMERICAN GAS ASSOCIATION, (A.G.A.)

A.G.A. Directory of Certified Appliances and Accessories

(Application for copies should be addressed to the American Gas Association, 8501 East Pleasant Valley Road, Cleveland, OH 44131.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 Standard product. The broiler-toaster-griddle delivered under this specification shall be the manufacturer's standard commercial product, except for any changes specified herein. All broiler-toaster-griddles furnished under this specification and any one contract including parts and assemblies thereof, shall be identical.

* 3.2 Codes and standards. The broiler-toaster-griddle shall comply with the applicable requirements of ANSI No. Z21.3 and NSF Standard No. 4.

* 3.2.1 Compliance. Prior to approval of the initial production lot, the contractor shall submit to the contracting officer, or his authorized representative, satisfactory evidence that the broiler-toaster-griddle he proposes to furnish under this specification complies with the applicable requirements of ANSI Z21.3 and NSF Standard No. 4 as follows:

* 3.2.1.1 ANSI. Acceptable evidence of meeting the applicable requirements of ANSI Z21.3 shall be a reproduced copy of the American Gas Association (A.G.A.) Appliance Certificate, a listing in the latest A.G.A. "Directory of Certified Appliances and Accessories", or a certified test report from a nationally recognized independent testing laboratory acceptable to the Government indicating compliance with ANSI Z21.3.

* 3.2.1.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished broiler-toaster-griddle, or

(2) A certification for the broiler-toaster-griddles issued by NSF under their special one-time evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the broiler-toaster-griddles are being procured, indicating that they have been tested and conform to NSF Standard No. 4.

* 3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer for broiler-toaster-griddles, provided the finished item complies with all provisions of this specification. Commercial tolerances shall apply to all metal thicknesses and diameters.

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3.3.1 Stainless steel sheet and strip. Stainless steel sheet and strip shall conform to any type in the 300 series of ASTM A 167 or any type in the 400 series of ASTM A 176.

3.4 Design. The broiler-toaster-griddle shall be a floor-model design, consisting essentially of a broiler, surmounted by a griddle, all constructed as a single unit. The broiler-toaster-griddle shall be of the concealed manifold type and shall be designed so that repairs or replacements can be made from the front or top without major disassembly of the item.

3.4.1 Burner capacity. Gas burners shall be located to heat the griddle and the broiler simultaneously. Gas burners shall have the capacity to maintain at least 75 British thermal units (Btu) per square inch of griddle cooking surface, and have a maximum installed capacity of not more than 80,000 Btu per hour when tested as specified in 4.4.2.

* 3.4.2 Fuel system. The fuel system, including burners shall be designed to efficiently burn natural gas and liquified petroleum gas, without gas leakage (see 4.4.1). Burner gas valves and pilot gas valves shall be of the fixed orifice type with interchangeable orifice hoods. Each valve shall be furnished with two orifice hoods or bushings. One orifice hood or bushing drilled for natural gas and the other drilled for liquified petroleum gas. The contractor shall provide instructions for conversion from one type of gas to the other. The orifice hoods shall be color coded as follows: (1) unpainted brass for natural gas; (2) black for liquified petroleum gas.

3.5 Construction.

3.5.1 Dimensions. The dimensions of the broiler-toaster-griddle shall be as follows:

	Inches
Height including legs (from floor to griddle top)	36
Width (side to side)	24 to 32
Depth (front to rear)	36 to 42

3.5.2 Body. All metal parts of both grade A and grade B broiler-toaster-griddles shall be carbon steel, except the body exterior front, sides, and back, of the grade A broiler-toaster-griddles shall be stainless steel specified in 3.4.1, with a 2B finish or better.

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3.5.3 Griddle plate. The griddle plate shall be one piece of cast iron, cast steel, or hot-rolled steel plate at least 800 square inches in area and not more than 3/4 inches thick. The finish of the top of the plate shall be not coarser than 125-micro inches in accordance with ANSI B46.1. A rear back-splash, and either straight or tapered sides shall be furnished.

3.5.4 Grease receptable. A grease trough with drain hole shall be provided to drain grease from the griddle-plate into a removable grease receptacle. The grease receptacle shall have a minimum capacity of 1 quart, and shall be easily removable from the front of the broiler-toaster-griddle.

3.5.5 Broiler grid. The broiler grid shall be the spring-balanced type, at least 500 square inches in area. The grid shall be easily removable and shall be adjustable vertically by means of a lifting mechanism having a conveniently located and easily operated handle. The handle shall be so located that at no time during cooking operations shall its temperature exceed 140°F. when tested as specified in 4.4.1. The broiler shall be provided with a removable drip pan at the bottom of the broiler compartment.

3.5.6 Legs. The broiler-toaster-griddle shall have four legs with a 2-inch adjustment. Height of legs shall conform to NSF Standard No. 4.

3.6 Finish. The broiler-toaster-griddle shall be clean, with no foreign matter imbedded in the finish. The finish shall have no scratches, gouges, pits, chips, grit, rust or scale. All parts of both grade A and grade B broiler-toaster-griddles, including broiler interiors, shall have the manufacturers' standard finish, except as specified in 3.5.3. Porcelain enameled surfaces shall be smooth and uniform, without runs, drips, or areas of thin film or no film, and with a minimum fired thickness of 13 mils.

* 3.7 Marking. Identification and, if applicable, operating instructions shall be permanently and legibly marked directly on the item or on an aluminum, brass, or stainless steel plate firmly affixed to the item. Marking shall be stamped, embossed, engraved or photosensitive. The plate thickness shall be not less than 0.012 inch.

* 3.7.1 Identification. Each broiler-toaster-griddle identification plate shall include the manufacturer's model and serial number, name, trade name or trademark, of such known character as to be readily identifiable to the manufacturer. In addition, such information as required by ANSI Z21.3.

3.8 Operation. The broiler-toaster-griddle burners and burner controls shall operate satisfactorily and each moving part shall fit and operate smoothly when tested as specified in 4.4.1.

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3.9 Workmanship. The finished broiler-toaster-griddle shall be free from defects, marred surfaces, dents, punctures, fractures, rough edges, burrs, and slivers. Fastener threads shall not be broken or stripped. All fasteners shall be tight. No component shall be missing, or malformed. All components shall be carefully assembled, adjusted, and aligned.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The inspection lot shall be the number of broiler-toaster-griddles of one grade offered for inspection at one time. The sample unit for the inspection shall be one completely assembled broiler-toaster-griddle.

* 4.2.2.1 Visual examination. Examination shall be made of the completely fabricated broiler-toaster-griddle for defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I. Classification of defects

Examine	Defect	Classification Major	Classification Minor
Finish	Not as specified	X	
	Runs, sags, drips, or area of no film or thin film in enameled surfaces		X
Design, construction and workmanship	Grid mechanism not spring balanced type	X	
	Any part missing	X	
	Any part fractured, split, punctured, sprung, or malformed	X	
	Any scratches, gouges, pits, chips, grit, rust or scale	X	
	Pourous sections, cracks, or dishouts in surface of griddle plate	X	
	Component that does not function as specified	X	
	Adjustable assembly that cannot be adjusted or aligned	X	
	Components that require servicing are inaccessible	X	
	Hoods not color coded		X
	Two hoods (unpainted brass and black) per burner valve not provided	X	
	Exterior front, sides, and back of grade A range not corrosion-resisting steel	X	
	Back splash not furnished	X	
	Not furnished with a removable drip pan	X	
	Marred surfaces, dents, fractures, rough edges, burrs, slivers	X	
	Loose fasteners; threaded fasterners stripped or broken		X
Marking	Missing, incomplete, not legible	X	

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4.2.2.2 Dimensional examination. Inspection shall be made of the end item to determine compliance with this specification. Any dimension, including the capacity of the grease receptacle and the thickness of enamel coatings, not within the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 6.5, expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item. The tests specified in 4.4.1 and 4.4.2 shall be performed on each completely assembled broiler-toaster-griddle. Failure to pass the tests shall be cause for rejection of the item.

* 4.2.3 Inspection of preparation for delivery. Preservation - packaging, packing and marking shall be inspected in accordance with the quality assurance provisions in MIL-R-11337.

* 4.3 Certification examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Test.

4.4.1 Operation test. The assembled broiler-toaster-griddle shall be connected to a source of fuel and shall be tested to determine compliance with the gas-tightness requirements of 3.4.2, the handle temperature requirements of 3.5.5, the operation of burners and controls, and the fit and smooth operation of all moving parts, as specified in 3.7. Handle temperature shall be taken after the burners have been operating at the highest temperature for one hour. The other parts of the test shall be made prior to lighting the burners and after the burners have been operating at the highest operating temperature for one hour. Either of the gases specified in 3.4.2 shall be used for the test. Any nonconformance shall constitute failure of this test.

* 4.4.2 Performance test. Using the same gas source as in 4:4.1, preheat the broiler in accordance with manufacturers' standard procedure. Check for Btu requirements specified in 3.4.1 as follows: Connect a gas volume meter into the gas supply line; operate burner at rated Btu setting; operate burner for one-half hour; and using the heating value of the gas supplied, compute the input in Btu/hr. Any nonconformance shall constitute failure of this test.

* 5. PACKAGING

5.1 Preservation - packaging. Preservation - packaging shall be level A or C as specified (see 6.2) in accordance with the applicable requirements of MIL-R-11337.

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* 5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Each broiler-toaster-griddle preserved and packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-R-11337.

* 5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with the applicable requirements of MIL-R-11337.

6. NOTES

6.1 Intended use. The broiler-toaster-griddle covered by this specification is intended for use in military kitchens and galleys for broiling, toasting, and griddling of foods.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Grade of broiler-toaster-griddle required (see 1.2).
(c) Selection of applicable levels of preservation - packaging, and packing required (see 5.1 and 5.2).

6.3 Contract data requirements. Any requirements for equipment manuals and drawings for the item covered by this specification should be included in DD Form 1423, Contract Data Requirements List, and cited in the contract.

6.4 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Army - MD
DSA - GS

Preparing activity:

Army - GL

Project No. 7310-0355

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-B-40119D BROILER-TOASTER-GRIDDLES, GAS FIRED.

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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